Result 1 (graded left identity). [Topology foundations, graded map left identity]
Let K be a type. Let *field* be a field structure on K. Let M and N be elements of GrMod_K. Let f be a degree-preserving map from M to N. Then

$$(f \circ \mathrm{id}_M) = f.$$

Result 2 (graded right identity). [Topology foundations, graded map right identity]
Let K be a type. Let *field* be a field structure on K. Let M and N be elements of GrMod_K. Let f be a degree-preserving map from M to N. Then

$$(\mathrm{id}_N \circ f) = f.$$

Result 3 (graded associativity). [Topology foundations, graded map associativity]
Let K be a type. Let *field* be a field structure on K. Let M, N, P and Q be elements of GrMod_K. Let f be a degree-preserving map from M to N. Let g be a degree-preserving map from N to P. Let h be a degree-preserving map from P to Q. Then

$$(h \circ (g \circ f)) = ((h \circ g) \circ f).$$

LLM-assisted example and proof (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.